As real-time capturing stops, the predicted names of the images stored in the list that has been 'outputted' are then displayed on the dashboard of the attendance management system.

Results

The images of students in the student database that are used as the training data for the algorithm can be seen in Figure 8.

When we run the code, the real-time capturing of the faces of the students starts. Now, when a person comes in front of the camera, the system detects the face and displays the name of the person if the image of the person is in the student database; otherwise, it displays the message, 'Unknown'.

For real-time capture, we have used the PlayStation eye camera which is capable of capturing standard video with frame rates of 60Hz at a 640×480-pixel resolution, and 120Hz at 320×240 pixel resolution.

We have repeated these results multiple times to check the accuracy and, every time, the application is able to identify the person's credentials correctly.

To summarise, we have implemented an attendance system using facial recognition, which marks the presence of the students in a class automatically and with no manual effort.

This can be a Web development project, for maintaining training data in an online database. Facial recognition can also be used to identify criminals, if sufficient information is available. END

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